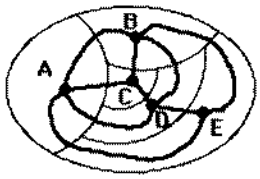
SECTION 10.8 Graph Coloring

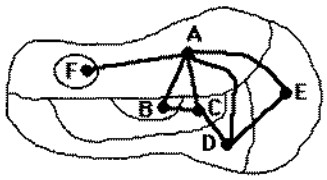
Like the problem of finding Hamilton paths, the problem of finding colorings with the fewest possible colors probably has no good algorithm for its solution. In working these exercises, for the most part you should proceed by trial and error, using whatever insight you can gain by staring at the graph (for instance, finding large complete subgraphs). There are also some interesting exercises here on coloring the edges of graphs—see Exercises 21–26. Exercises 29–31 are worth looking at, as well: they deal with a fast algorithm for coloring a graph that is not guaranteed to produce an optimal coloring.

1. We construct the dual graph by putting a vertex inside each region (but not in the unbounded region), and drawing an edge between two vertices if the regions share a common border. The easiest way to do this is illustrated in our answer. First we draw the map, then we put a vertex inside each region and make the connections. The dual graph, then, is the graph with heavy lines.



The number of colors needed to color this map is the same as the number of colors needed to color the dual graph. Since A, B, C, and D are mutually adjacent, at least four colors are needed. We can color each of the vertices (i.e., regions) A, B, C, and D a different color, and we can give E the same color as we give C.

3. We construct the dual as in Exercise 1.



As in Exercise 1, the number of colors needed to color this map is the same as the number of colors needed to color the dual graph. Three colors are clearly necessary, because of the triangle ABC, for instance. Furthermore three colors suffice, since we can color vertex (region) A red, vertices B, D, and F blue, and vertices C and E green.

5. For Exercises 5–11, in order to prove that the chromatic number is k, we need to find a k-coloring and to show that (at least) k colors are needed. Here, since there is a triangle, at least 3 colors are needed. Clearly 3 colors suffice, since we can color a and d the same color.

7. Since there is a triangle, at least 3 colors are needed. Clearly 3 colors suffice, since we can color a and c the same color.

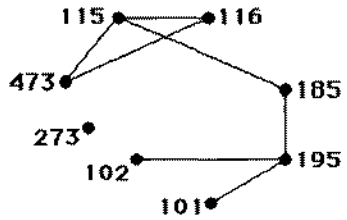
9. Since there is an edge, at least 2 colors are needed. The coloring in which b, d, and e are red and a and c blue shows that 2 colors suffice.

11. Since there is a triangle, at least 3 colors are needed. It is not hard to construct a 3-coloring. We can let a, f, h, j, and n be blue; let b, d, g, k, and m be green; and let c, e, i, l, and o be yellow.

13. If a graph has an edge (not a loop, since we are assuming that the graphs in this section are simple), then its chromatic number is at least 2. Conversely, if there are no edges, then the coloring in which every vertex receives the same color is proper. Therefore a graph has chromatic number 1 if and only if it has no edges.

15. In Example 4 we saw that the chromatic number of C_n is 2 if n is even and 3 if n is odd. Since the wheel W_n is just C_n with one more vertex, adjacent to all the vertices of the C_n along the rim of the wheel, W_n clearly needs exactly one more color than C_n (for that middle vertex). Therefore the chromatic number of W_n is 3 if n is even and 4 if n is odd.

17. Consider the graph representing this problem. The vertices are the 8 courses, and two courses are joined by an edge if there are students taking both of them. Thus there are edges between every pair of vertices except the 7 pairs listed. It is much easier to draw the complement than to draw this graph itself; it is shown below.



We want to find the chromatic number of the graph whose complement we have drawn; the colors will be the time periods for the exams. First note that since Math 185 and the four CS courses form a K_5 (in other words, there are no edges between any two of these in our picture), the chromatic number is at least 5. To show that it equals 5, we just need to color the other three vertices. A little trial and error shows that we can make Math 195 the same color as (i.e., have its final exam at the same time as) CS 101; and we can make Math 115 and 116 the same color as CS 473. Therefore five time slots (colors) are sufficient.

19. We model the problem with the intersection graph of these sets. Note that every pair of these intersect except for C_4 and C_5. Thus the graph is K_6 with that one edge deleted. Clearly its chromatic number is 5, since we need to color all the vertices different colors, except that C_4 and C_5 may have the same color. In other words, 5 meeting times are needed, since only committees C_4 and C_5 can meet simultaneously.

21. Note that the number of colors needed to color the edges is at least as large as the largest degree of a vertex, since the edges at each vertex must all be colored differently. Hence if we can find an edge coloring with that many colors, then we know we have found the answer. In Exercise 5 there is a vertex of degree 3, so the edge chromatic number is at least 3. On the other hand, we can color $\{a, c\}$ and $\{b, d\}$ the same color, so 3 colors suffice. In Exercise 6 the 6 edges incident to g must all get different colors. On the other hand, it is not hard to complete a proper edge coloring with only these colors (for example, color edge $\{a, f\}$ with the same color as used on $\{d, g\}$), so the answer is 6. In Exercise 7 the answer must be at least 3; it is 3 since edges that appear as parallel line segments in the picture can have the same color. In Exercise 8 clearly 4 colors are required, since the vertices have degree 4. In fact 4 colors are sufficient. Here is one proper 4-coloring (we denote edges in the obvious shorthand notation): color 1 for ac, be, and df; color 2 for ae, bd, and cf; color 3 for ab, cd, and ef; and color 4 for ad, bf, and ce. In Exercise 9 the answer must be at least 3; it is easy to construct a 3-coloring of the edges by inspection: $\{a, b\}$ and $\{c, e\}$ have the same color, $\{a, d\}$ and $\{b, c\}$ have the same color, and $\{a, e\}$ and $\{c, d\}$ have the same color. In Exercise 10 the largest degree is 6 (vertex i has degree 6); therefore at least 6 colors are required. By trial and error we come up with this coloring using 6 colors (we use the obvious shorthand notation for edges); there are many others, of course. Assign color 1 to ag, cd, and hi; color 2 to ab, cf, dg, and ei; color 3 to bh, cg, di, and ef; color 4 to ah, ci, and de; color 5 to bi, ch, and fg; and color 6 to ai, bc, and gh. Finally, in Exercise 11 it is easy to construct an edge-coloring with 4 colors; again the edge chromatic number is the maximum degree of a vertex.

Despite the appearances of these examples, it is not the case that the edge chromatic number of a graph is always equal to the maximum degree of the vertices in the graph. The simplest example in which this is not

true is K_3. Clearly its edge chromatic number is 3 (since all three edges are adjacent to each other), but its maximum degree is 2. There is a theorem, however, stating that the edge chromatic number is always equal to either the maximum degree or one more than the maximum degree.

23. **a)** The n-cycle's edges are just like the n-cycle's vertices (each adjacent to the next as we go around the cycle), so the edge chromatic number is the same, namely 2 if n is even and 3 if n is odd, as in Example 4. **b)** The edge chromatic number is at least n, because the radial edges are all pairwise adjacent and therefore must all have distinct colors. Suppose we call these colors 1 through n proceeding clockwise. We need no additional colors for the edges of the cycle, because we can color the edge adjacent to the spokes colored 1 and 2 with color 3 and proceed clockwise with colors 4, 5, ..., $n-1$, n, 1, and 2. Therefore $\chi'(W_n) = n$.

25. Two edges that have the same color share no endpoints. Therefore if more than $n/2$ edges were colored the same, the graph would have more than $2(n/2) = n$ vertices.

27. This problem can be modeled with the intersection graph of the sets of steps during which the variables must be stored. This graph has 7 vertices, t through z; there is an edge between two vertices if the two variables they represent must be stored during some common step. The answer to the problem is the chromatic number of this graph. Rather than considering this graph, we look at its complement (it has a lot fewer edges). Here two vertices are adjacent if the sets (of steps) do not intersect. The only edges are $\{u,w\}$, $\{u,x\}$, $\{u,y\}$, $\{u,z\}$, $\{v,x\}$, $\{x,z\}$. Note that there are no edges in the complement joining any two of $\{t,v,w,y,z\}$, so that these vertices form a K_5 in the original graph. Thus the chromatic number of the original graph is at least 5. To see that it is 5, note that vertex u can have the same color as w, and x can have the same color as z (these pairs appear as edges in the complement). Since the chromatic number is 5, we need 5 registers, with variables u and w sharing a register, and vertices x and z sharing one.

29. First we need to list the vertices in decreasing order of degree. This ordering is not unique, of course; we will pick e,a,b,c,f,h,i,d,g,j. Next we assign color 1 to e, and then to f and d, in that order. Now we assign color 2 to a, c, i, and g, in that order. Finally, we assign color 3 to b, h and j, in that order. Thus the algorithm gives a 3-coloring. Since the graph contains triangles, we know that this is the best possible, so the algorithm "worked" here (but it need not always work—see Exercise 27).

31. A simple example in which the algorithm may fail to provide a coloring with the minimum number of colors is C_6, which of course has chromatic number 2. Since all the vertices are of degree 2, we may order them v_1, v_4, v_2, v_3, v_5, v_6, where the edges are $\{v_1,v_2\}$, $\{v_2,v_3\}$, $\{v_3,v_4\}$, $\{v_4,v_5\}$, $\{v_5,v_6\}$, and $\{v_1,v_6\}$. Then v_1 gets color 1, as does v_4. Next v_2 and v_5 get color 2; and then v_3 and v_6 must get color 3.

33. We need to show that the wheel W_n when n is an odd integer greater than 1 can be colored with four colors, but that any graph obtained from it by removing one edge can be colored with three colors. Four colors are needed to color this graph, because three colors are needed for the rim (see Example 4), and the center vertex, being adjacent to all the rim vertices, will require a fourth color. To complete the proof that W_n is chromatically 4-critical, we must show that the graph obtained from W_n by deleting one edge can be colored with three colors. There are two cases. If we remove a rim edge, then we can color the rim with two colors, by starting at an endpoint of the removed edge and using the colors alternately around the portion of the rim that remains. The third color is then assigned to the center vertex. On the other hand, if we remove a spoke edge, then we can color the rim by assigning color #1 to the rim endpoint of the removed edge and colors #2 and #3 alternately to the remaining vertices on the rim, and then assign color #1 to the center.

35. We give a proof by contradiction. Suppose that G is chromatically k-critical but has a vertex v of degree $k-2$ or less. Remove from G one of the edges incident to v. By definition of "k-critical," the resulting graph can be colored with $k-1$ colors. Now restore the missing edge and use this coloring for all vertices except v. Because we had a proper coloring of the smaller graph, no two adjacent vertices have the same color. Furthermore, v has at most $k-2$ neighbors, so we can color v with an unused color to obtain a proper $(k-1)$-coloring of G. This contradicts the fact that G has chromatic number k. Therefore our assumption was wrong, and every vertex of G must have degree at least $k-1$.

37. **a)** Note that vertices d, e, and f are mutually adjacent. Therefore six different colors are needed in a 2-tuple coloring, since each of these three vertices needs a disjoint set of two colors. In fact it is easy to give a coloring with just six colors: Color a, d, and g with $\{1,2\}$; color c and e with $\{3,4\}$; and color b and f with $\{5,6\}$. Thus $\chi_2(G) = 6$.

b) This one is trickier than part **(a)**. There is no coloring with just six colors, since if there were, we would be forced (without loss of generality) to color d with $\{1,2\}$; e with $\{3,4\}$; f with $\{5,6\}$; then g with $\{1,2\}$, b with $\{5,6\}$, and c with $\{3,4\}$. This gives no free colors for vertex a. Now this may make it appear that eight colors are required, but a little trial and error shows us that seven suffice: Color a with $\{2,4\}$; color b and f with $\{5,6\}$; color d with $\{1,2\}$; color c with $\{3,7\}$; color e with $\{3,4\}$; and color g with $\{1,7\}$. Thus $\chi_2(H) = 7$.

c) This is similar to part **(a)**. Here nine colors are necessary and sufficient, since a, d, and g can get one set of three colors; b and f can get a second set; and c and e can get a third set. Clearly nine colors are necessary to color the triangles.

d) First we construct a coloring with 11 colors: Color a with $\{3,6,11\}$; color b and f with $\{7,8,9\}$; color d with $\{1,2,3\}$; color c with $\{4,5,10\}$; color e with $\{4,6,11\}$; and color g with $\{1,2,5\}$. To prove that $\chi_3(H) = 11$, we must show that it is impossible to give a 3-tuple coloring with only ten colors. If such a coloring were possible, without loss of generality we could color d with $\{1,2,3\}$, e with $\{4,5,6\}$, f with $\{7,8,9\}$, and g with $\{1,2,10\}$. Now nine colors are needed for the three vertices a, b, and c, since they form a triangle; but colors 1 and 2 are already used in vertices adjacent to all three of them. Therefore at least $9+2 = 11$ colors are necessary.

39. The frequencies are the colors, the zones are the vertices, and two zones that are so close that interference would be a problem are joined by an edge in the graph. Then it is clear that a k-tuple coloring is exactly an assignment of frequencies that avoids possible interference.

41. We use induction on the number of vertices of the graph. Every graph with five or fewer vertices can be colored with five or fewer colors, since each vertex can get a different color. That takes care of the basis case(s). So we assume that all graphs with k vertices can be 5-colored and consider a graph G with $k+1$ vertices. By Corollary 2 in Section 10.7, G has a vertex v with degree at most 5. Remove v to form the graph G'. Since G' has only k vertices, we 5-color it by the inductive hypothesis. If the neighbors of v do not use all five colors, then we can 5-color G by assigning to v a color not used by any of its neighbors. The difficulty arises if v has five neighbors, and each has a different color in the 5-coloring of G'. Suppose that the neighbors of v, when considered in clockwise order around v, are a, b, c, m, and p. (This order is determined by the clockwise order of the curves representing the edges incident to v.) Suppose that the colors of the neighbors are azure, blue, chartreuse, magenta, and purple, respectively. Consider the azure-chartreuse subgraph (i.e., the vertices in G colored azure or chartreuse and all the edges between them). If a and c are not in the same component of this graph, then in the component containing a we can interchange these two colors (make the azure vertices chartreuse and vice versa), and G' will still be properly colored. That makes a chartreuse, so we can now color v azure, and G has been properly colored. If a and c are in the same component, then there is a path of vertices alternately colored azure and chartreuse joining a and c. This path together with

edges av and vc divides the plane into two regions, with b in one of them and m in the other. If we now interchange blue and magenta on all the vertices in the same region as b, we will still have a proper coloring of G', but now blue is available for v. In this case, too, we have found a proper coloring of G. This completes the inductive step, and the theorem is proved.

43. We follow the hint. Because the measures of the interior angles of a pentagon total 540°, there cannot be as many as three interior angles of measure more than 180° (reflex angles). If there are no reflex angles, then the pentagon is convex, and a guard placed at any vertex can see all points. If there is one reflex angle, then the pentagon must look essentially like figure (a) below, and a guard at vertex v can see all points. If there are two reflex angles, then they can be adjacent or nonadjacent (figures (b) and (c)); in either case, a guard at vertex v can see all points. (In figure (c), choose the reflex vertex closer to the bottom side.) Thus for all pentagons, one guard suffices, so $g(5) = 1$.



45. The figure suggested in the hint (generalized to have k prongs for any $k \geq 1$) has $3k$ vertices. Consider the set of points from which a guard can see the tip of the first prong, the set of points from which a guard can see the tip of the second prong, and so on. These are disjoint triangles (together with their interiors). Therefore a separate guard is needed for each of the k prongs, so at least k guards are needed. This shows that $g(3k) \geq k = \lfloor 3k/3 \rfloor$. To handle values of n that are not multiples of 3, let $n = 3k + i$, where $i = 1$ or 2. Then obviously $g(n) \geq g(3k) \geq k = \lfloor n/3 \rfloor$.

GUIDE TO REVIEW QUESTIONS FOR CHAPTER 10

1. **a)** See pp. 641–642 and Table 1 in Section 10. **b)** See Exercise 1 in Section 10.1.

2. See all the examples Section 10.1.

3. See Theorem 1 in Section 10.2.

4. See Theorem 2 in Section 10.2.

5. See Theorem 3 in Section 10.2.

6. **a)** See Example 5 in Section 10.2. **b)** See Example 13 in Section 10.2.
c) See Example 6 in Section 10.2. **d)** See Example 7 in Section 10.2.
e) See Example 8 in Section 10.2.

7. **a)** n, $C(n,2)$ **b)** $m+n$, mn **c)** n, n **d)** $n+1$, $2n$ **e)** 2^n, $n2^{n-1}$

8. **a)** See p. 656. **b)** K_2 and C_{2m}
c) (See also Example 12 and Exercise 66 in Section 10.2.) The following algorithm is an efficient way to determine whether a connected graph can be 2-colored (which is the same thing as saying that it is bipartite); apply it to each component of the given graph. First color any vertex red. Then color all vertices adjacent to this vertex blue. Then look at all vertices adjacent to these just-colored blue vertices. If any of them are already colored blue, then stop and declare the graph not to be bipartite; otherwise color all the uncolored ones red. Next look at all vertices adjacent to all the vertices just colored red. If any of them are already

colored red, then stop and declare the graph not to be bipartite; otherwise color all the uncolored ones blue. Continue in this way until no more vertices can be colored. If we get this far, then (this component of) the graph is bipartite. (If uncolored vertices remain, they are in a different component, so we can repeat the entire process starting with any uncolored vertex.)

9. **a)** adjacency lists, adjacency matrices, incidence matrices

b) Look at Figure 1 in Section 10.3, and let G be the graph consisting of the vertices and edges shown there, together with edges $\{b, c\}$ and $\{b, e\}$. Its adjacency lists are shown on p. 668 (Table 1), once we add c and e to the list of adjacent vertices of b, and add b to the list for c and for e. Its adjacency matrix and incidence matrix are as follows (using alphabetical order):

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

10. **a)** See p. 672.

b) See p. 672; number of vertices, number of edges, degrees of vertices, existence of triangle (C_3 as subgraph), and existence of Hamilton circuit are all invariants.

c) See Figure 10 in Section 10.3. **d)** no

11. **a)** See p. 681. **b)** See p. 682.

12. **a)** See p. 669. **b)** See p. 673. **c)** See Theorem 2 in Section 10.4.

13. **a)** See p. 694. **b)** See pp. 693–694.

c) All edges are in the same component and there are at most two vertices of odd degree (see Theorems 1 and 2 in Section 10.5).

d) All edges are in the same component and there are no vertices of odd degree (see Theorem 1 in Section 10.5).

14. **a)** See p. 698. **b)** the existence of a cut edge or a cut vertex

15. finding the shortest highway route between two cities; finding the cheapest way to lease telephone lines to join two communications centers, using intermediate switching centers

16. **a)** See pp. 709–712. **b)** See Exercise 5, third part, in Section 10.6.

17. **a)** See p. 719. **b)** K_6

18. **a)** If the planar graph has v vertices, e edges, and c components, and is embedded in the plane so as to form r regions, then $v - e + r = 1 + c$. (The theorem as stated in the text—Theorem 1 in Section 10.7—assumes that $c = 1$, but this is the more general statement.)

b) By Corollary 1 in Section 10.7, for every planar graph with at least three vertices we know that $e \leq 3v - 6$ (connectivity need not be assumed). Thus we can show that a graph is nonplanar by showing that it has too many edges, namely more than $3v - 6$ edges. For example, K_6 is nonplanar, since it has 15 edges, and $15 \not\leq 3 \cdot 6 - 6$.

19. See Theorem 2 in Section 10.7. A graph is planar if and only if it does not contain a subgraph homeomorphic to K_5 or $K_{3,3}$.

20. **a)** See p. 728. **b)** n **c)** 2 if n is even, 3 if n is odd **d)** 2

21. Every planar graph can be 4-colored (i.e., has chromatic number at most 4), but K_6, for instance, requires six colors.

22. See Examples 5–7 in Section 10.8.

SUPPLEMENTARY EXERCISES FOR CHAPTER 10

1. Every vertex has degree 50. Thus the sum of the degrees must be $50 \cdot 100 = 5000$. By the handshaking theorem, the graph therefore has $5000/2 = 2500$ edges.

3. Both graphs have a lot of symmetry to them, and the degrees of the vertices are the same, so we might hope that they are isomorphic. Let us try to form the correspondence f. First note that there is a 4-cycle u_1, u_5, u_2, u_6, u_1 in the first graph. Suppose that we try letting it correspond to the 4-cycle v_1, v_2, v_3, v_4, v_1 in the second graph. Thus we let $f(u_1) = v_1$, $f(u_5) = v_2$, $f(u_2) = v_3$, and $f(u_6) = v_4$. The rest of the assignments are forced; since u_7 is the other vertex adjacent to u_1, we must let $f(u_7) = v_6$, since v_6 is the other vertex adjacent to v_1 (which is $f(u_1)$). Similarly, $f(u_3) = v_7$, $f(u_8) = v_8$, and $f(u_4) = v_5$. Now we just have to check that the vertices corresponding to the vertices in the 4-cycle v_5, v_6, v_7, v_8, v_5 in the second graph form a 4-cycle in that order. Since these vertices form the 4-cycle u_4, u_7, u_3, u_8, u_4, our correspondence works.

5. These graphs are isomorphic, although the isomorphism is hard to find. One approach that can lead to the isomorphism is to draw the complement of each graph. The complements are a little simpler than the original graphs, since they have fewer edges. When we do this, it is easy to give a planar representation of each. Then by looking at the sizes of the regions we can find an isomorphism. One such correspondence is $u_1 \leftrightarrow v_5$, $u_2 \leftrightarrow v_4$, $u_3 \leftrightarrow v_7$, $u_4 \leftrightarrow v_3$, $u_5 \leftrightarrow v_8$, $u_6 \leftrightarrow v_2$, $u_7 \leftrightarrow v_6$, and $u_8 \leftrightarrow v_1$. We just need to check that all the edges are preserved by this correspondence.

7. It follows immediately from the definition that the complete m-partite graph with parts $n_1, n_2, \ldots, n_m$ has $n_1 + n_2 + \cdots + n_m = \sum_{i=1}^{m} n_i$ vertices. We will organize the count of the edges by looking at which parts the edges join. Fix $1 \leq i < j \leq m$, and consider the edges between the ith part and the jth part. It is easy to see from the product rule that there are $n_i n_j$ edges. Therefore to get all the edges, we have to add all these products, for all possible pairs (i, j). Thus the number of edges is $\sum_{1 \leq i < j \leq m} n_i n_j$.

9. **a)** We prove that each set is a subset of the other. If $x \in N(A \cup B)$, then x is adjacent to some vertex $v \in A \cup B$. If $v \in A$, then $x \in N(A)$ and therefore also in $N(A) \cup N(B)$; the other case is similar. Conversely, if $x \in N(A) \cup N(B)$, then without loss of generality assume $x \in N(A)$. That means that x is adjacent to some vertex $v \in A$. Since $v \in A \cup B$, we have $x \in N(A \cup B)$.

b) If $x \in N(A \cap B)$, then x is adjacent to some vertex $v \in A \cap B$. Since both $v \in A$ and $v \in B$, it follows that $x \in N(A)$ and $x \in N(B)$, whence $x \in N(A) \cap N(B)$. For an example to show that equality may fail, consider $K_{1,2}$, let A be the set consisting of one of the vertices of degree 1, and let B be the set consisting of the other vertex of degree 1. Then $A \cap B = \emptyset$, so $N(A \cap B) = \emptyset$, but $N(A) \cap N(B)$ is the vertex of degree 2.

11. There are many possible SDR's here, which can be found by trial and error. One is (m, e, p, x, n, a).

13. **a)** There are actually 15 subsets of $\{1, 2, 3, 4\}$ to check here (letting $I = \emptyset$ always gives the trivial true fact that $0 \leq 0$), but since each set S_i has at least three elements, the inequality is automatically true for all I of size less than 4. The only other case is $I = \{1, 2, 3, 4\}$ in which case both sides of the inequality are $|\{a, b, c, d\}|$.

b) Many choices are possible, such as (a, b, d, c).

15. Consider each of the ordered triples of vertices (u, v, w) of G. It may happen that u and v are neighbors and that v and w are neighbors; for all triples for which this is so, we want to know how many also have u adjacent to w. The corresponding fraction is by definition the clustering coefficient of G. Each triangle in G gives rise to 6 ordered triples that contribute to the numerator of this fraction. Each path of length 2 in G gives rise to one ordered triple that contribute to the denominator of this fraction. Thus $C(G)$ equals 6 times the number of triangles divided by the number of paths of length 2. For example, in K_3, there is one triangle and there are 6 paths of length 2, so $C(K_3) = 6 \cdot 1/6 = 1$. Similarly, $C(W_4) = 6 \cdot 4/36 = 2/3$.

17. Each of these can be phrased in terms of a probability—how likely something is to happen. It is a little subtle, though. In each case, we have to start with the common neighbor. See part **(a)** for the more precise wording; the others are similar.

a) How likely is it that two actors will have appeared in a film together if we know that there is a third actor who has appeared in a film with each of them? To be more precise, if we randomly pick an actor and two actors who have appeared in a film with that actor, how likely is it that those two actors have appeared in a film together. This is not the same as starting with two random actors who have a common co-star, because if there is more than one common co-star, the probability goes up.

b) How likely is it that two people who have a Facebook friend in common are themselves Facebook friends? A little more precisely: How likely is it that two of a randomly chosen person's friends are themselves friends?

c) How likely is it that two coauthors of a randomly chosen author are themselves coauthors?

d) How likely is it that two proteins that each interact with a randomly chosen protein interact with each other?

e) How likely is it that two routers each of which has a communications link to a randomly chosen router are themselves linked?

19. In general it is no easy task to find cliques. We need to be careful not to overlook things. We will denote a clique simply by listing the vertices in it, without punctuation. There is one K_4, namely $bcef$, and it is the largest clique. There are several K_3's not contained in this K_4, and they are all cliques: abg, adg, beg, and deg. Since every edge is contained in one of these five cliques (and there are no isolated vertices), there are no smaller cliques, so this list is complete.

21. See the comments for Exercise 19. We find the cliques by brute force and careful looking, hoping that we do not miss any. Some staring at the graph convinces us that there are no K_6's. There is one K_5, namely the clique $bcdjk$. There are two K_4's not contained in this K_5, which therefore are cliques: $abjk$, and $efgi$. All the K_3's not contained in any of the cliques listed so far are also cliques. We find abi, aij, bde, bei, bij, ghi, and hij. All the edges are in at least one of the cliques listed so far (and there are no isolated vertices), so we are done.

23. Clearly no single vertex is dominating by itself, but $\{c, d\}$ dominates, so it is a minimum dominating set (there are lots of others).

25. These graphs are quite a mess to draw (but see the answer section of the textbook for pictures), since they contain so many edges. Instead of drawing them, we describe them in set-theoretic terms. Note that we do not consider a queen on a square to control that square itself, since to do so would give us loops in the graph.

a) The vertex set consists of all pairs (i, j) with $1 \le i \le 3$ and $1 \le j \le 3$. Since a queen in any square controls all the squares in the same row and column, there are edges $\{(i, j), (i, j')\}$ and $\{(i, j), (i', j)\}$ for all i, j, i' and j' between 1 and 3 inclusive, with $i \ne i'$ and $j \ne j'$. These are not all, though, since we have to put in the diagonal controls. There are 10 such edges: $(1, 1)$, $(2, 2)$, and $(3, 3)$ are all joined to each other;